

Xianhao Carton Liu

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Education

Computer Science Ph.D. Student

Advisor: **Zhu-Tian Chen**

Research topic: Human AI Interaction in Physical Environments and Real World

University of Minnesota, Twin Cities

Aug 2024 – Now

B.Eng. with Honors in Artificial Intelligence (Turing Class)

Outstanding honored undergraduate in **Chu Kochen Honors College**

Zhejiang University

Sep 2020 – June 2024

Publications

Generative Proxy: Synthesizing Proxy-Based Interfaces for Real-World Interaction Across AR Glasses

Accepted in *UIST 2026* [Paper]

Jun 2026

- [Xianhao Carton Liu](#), William Chastek, Eric J Gonzalez, Mar Gonzalez-Franco, Chen Zhu-Tian

Can AR Embedded Visualizations Foster Appropriate Reliance on AI in Spatial Decision Making?

A Comparative Study of AR X-Ray vs. 2D Minimap

Accepted in *CHI 2026* [Paper] [Video] [Github]

Jan 2026

- [Xianhao Carton Liu](#), Difan Jia, Tongyu Nie, Evan Suma Rosenberg, Victoria Interrante, Chen Zhu-Tian

Reality Proxy: Fluid Interactions with Real-World Objects in MR via Abstract Representations

Accepted in *UIST 2025* [Paper] [Video]

May 2025

- Xiaolan Liu, Difan Jia, [Xianhao Carton Liu](#), Mar Gonzalez-Franco, Chen Zhu-Tian

Research Experience

Harvard Visual Computing Group, Harvard University

Research intern advised by [Zhu-Tian Chen](#) and [Hanspeter Pfister](#)

Cambridge, USA

Jan 2024 - May 2024

Stanford HCI Group, Stanford University

Research intern advised by [Jackie Yang](#) and [James Landay](#)

California, USA

Jul 2023 - Sep 2023

ZJU International Design Institute, Zhejiang University

Research intern advised by [Lingyun Sun](#) and [Zejian Li](#)

Hangzhou, China

Mar 2022 - Jun 2023

Projects

WebXR Fundamental 3D Interaction Methods for Object Manipulation

[Video] [Github]

- A WebXR interaction framework for object manipulation methods including translation, rotation and resizing.

WebXR for 3D interaction

Spring 2024

Generative XR Spatial Interfaces for Human Robot Co-exploration

[Video] [Github]

- Reconstructing and synthesizing real-world scene via mobile robot on real-time for human space interaction

Situated XR Interfaces

Summer 2025

Human-AI decision-making in spatial environment via AR visualizations

[Video] [Github]

- Understanding human overreliance and trust on AI in spatial decision-making via different AR visualizations

Human AI Collaboration

Spring 2025

Handbot——Apple WWDC2023 Swift Student Challenge

[Video] [Github]

- A swift playground encouraging critical thinking and editing on LLM and other genAI content in multi-modalities.

Human-centered GenAI

Apr 2023

Is raining threes worth it?——NBA 3-pointer Data Visualization

[Video] [Website]

- An interactive visualization website featuring NBA 3-pointers data and key players, received 17000+ views

Data Visualization

Dec 2022

Skills

Programming

HCI Methods

Tools

Agentic engineering/WebXR/Unity3D/Swift/ROS/Pytorch/ThreeJS/React/C#/C++/Python
Quantitative Analysis/Interview Thematic Analysis/Participatory Design and Analysis
Smart glasses and XR headsets/Embedded Computers and Wearable Devices/Turtlebot4/Design editors